Epochs, Batch, and Iterations in Deep Learning

What is an Epoch?

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An epoch refers to **one complete pass** through the entire training dataset once by the neural network. This includes:

- **One forward pass:** The model takes **inputs**, makes predictions, and computes errors.
- **One backward pass**: The model **adjusts its weights** based on the errors to improve future predictions.

Why are epochs important?

Training a model for a single epoch usually isn't enough to capture all the patterns in the data. By repeating the process for multiple epochs, the model gradually learns and improves its performance.

1 epoch = 1 complete iteration over the entire dataset, with each sample used once for both forward and backward passes.

For Example — Suppose you have a training dataset with **1,000 samples**, and you are training the model for **3 epochs**:

1 epoch: The model processes all **1,000 samples** once (forward + backward passes).

3 epochs: The model processes the **entire dataset 3 times**, meaning **3 forward and 3 backward passes** for each of the 1,000 samples.

Analogy:

Think of studying for an exam:

- **1 epoch**: You go through the entire syllabus once.
- **5 epochs**: You study the same syllabus five times, each time understanding more and solidifying your knowledge.

What is a Batch?

A **batch** refers to a **subset** of the training data that is processed together during **one forward pass** and **one backward pass** of the neural network. Since the entire dataset is often too large to process all at once, it is divided into smaller groups called **batches**.

What is Batch Size?

Batch size is a **hyperparameter** that defines the number of training examples included in one batch. It determines how much data is processed before the model updates its weights.

Example:

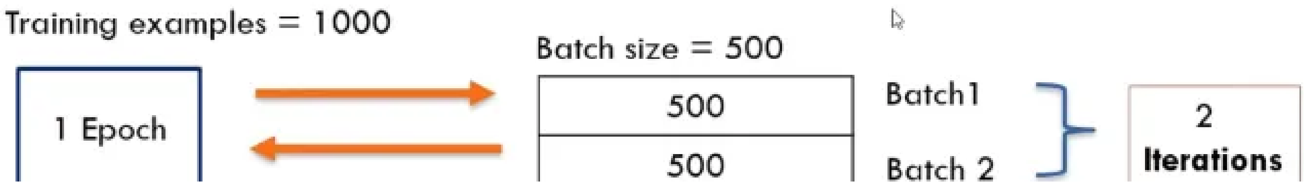
Suppose you have **1,000 training examples**, and you set the batch size to **500**

- The training data is divided into **2 batches**, each containing **500 examples.**
- To complete **1 epoch** (processing all 1,000 examples), the model will require **2 iterations:**

Batch 1: First 500 examples are processed.

Batch 2: Remaining 500 examples are processed.

This means the entire dataset is processed (1 epoch) in **2 iterations**.



Batch and Batch size

Iterations

An **iteration** refers to a single step of training where the model processes one **batch** of data and updates its parameters (weights and biases).

Each iteration includes two main parts:

1. **Forward Propagation**: The input data in the batch is passed through the network to generate predictions, and the loss (error) is calculated.
2. **Backward Propagation**: The gradients are calculated, and the model's parameters are updated to reduce the loss.

Relation Between Iterations and Epochs:

An **epoch** is completed when the entire dataset is processed once.

The **number of iterations in one epoch** depends on the **batch size** and the total number of training examples.

$$\text{Number of Iterations in One Epoch} = \frac{\text{Total Training Samples}}{}$$

Example

Suppose you have **1,000 training samples.** You set the **batch size** to **100.**

> *Here's how the iterations work*

The model will divide the training data into **10 batches** (1,000 ÷ 100 = 10).

Each batch will be processed once during one iteration.

After processing all 10 batches (10 iterations), the model completes **1 epoch.**

Analogy Example of an Iteration

- **The entire pizza** represents the full training dataset.
- **Each slice of pizza** represents a batch of the dataset.
- **Eating one slice** is one iteration, where you focus on just one part of the pizza (processing one batch of data).
- **Finishing the entire pizza** (all slices) is one epoch, meaning you have processed the entire dataset once.

If the pizza has **8 slices** and you eat **1 slice** at a time:

- You need **8 iterations** to finish 1 pizza (1 epoch).
- After finishing all slices, you can start on another pizza (another epoch) to get even more satisfaction (better model learning).

Why Are Iterations Important?

- **Efficient Training:** Training on smaller batches makes the process manageable and less memory-intensive compared to processing the entire dataset at once.
- **Progressive Updates:** Iterations allow the model to learn continuously by updating parameters after each batch, leading to faster optimization.
- **Scalability:** Iterations enable the handling of large datasets by dividing them into smaller, more manageable parts (batches).

Thanks for Reading!!

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